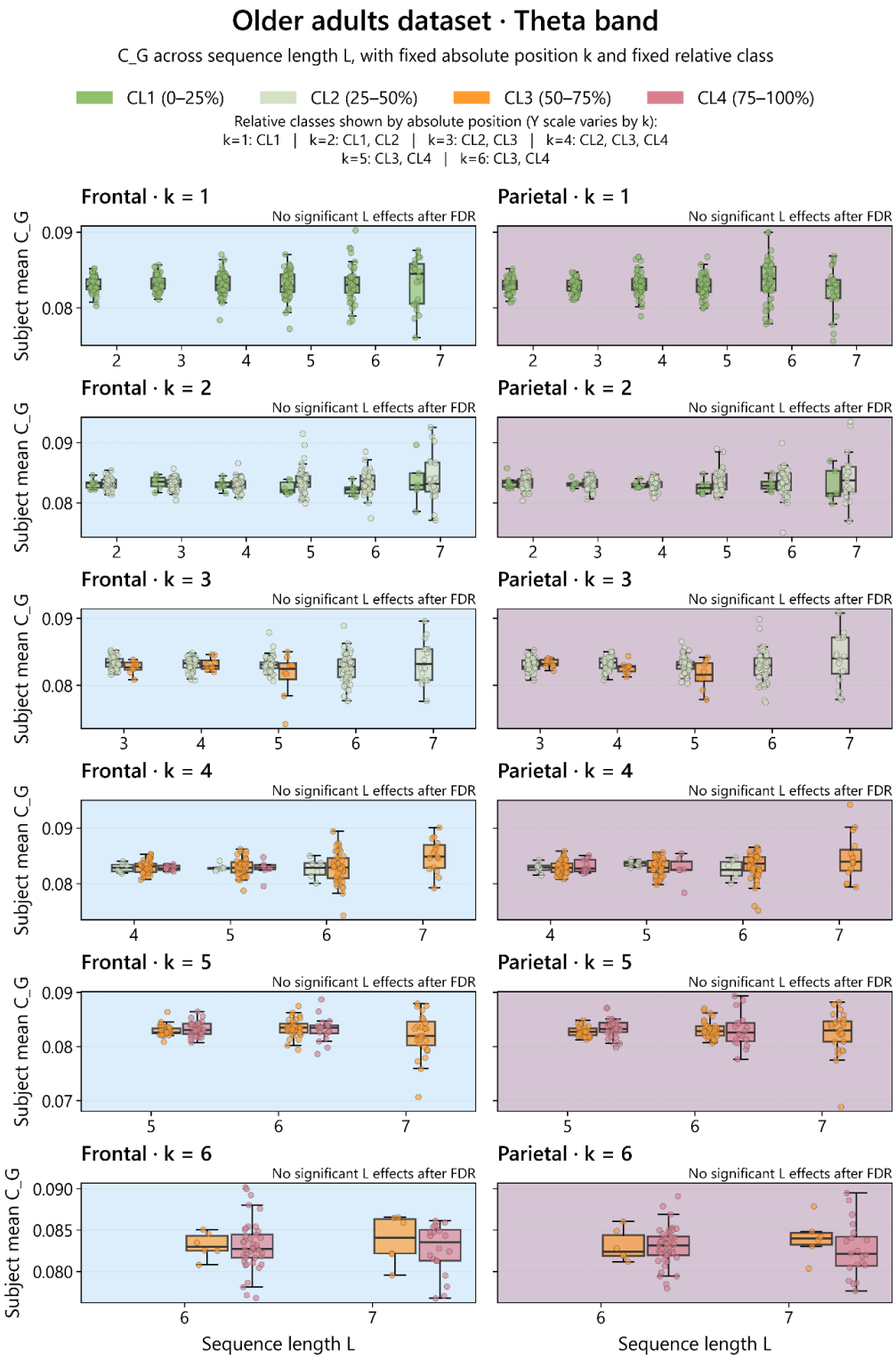


Fig. S11. Shannon graph complexity across sequence lengths in the older-adult dataset (theta band, 4–8 Hz). Rows represent absolute serial positions (k = 1–6); columns show frontal (left) and parietal (right) results. The horizontal axis indicates sequence length (L), and colors identify relative classes CL1–CL4, defined using each participant's Corsi span. Each point represents a participant's mean C_G computed from clean epochs belonging to correctly completed trials. Boxes indicate the interquartile range and central lines the median. Vertical scales are shared within rows and vary between rows. Sequence lengths were compared within each fixed k and class using paired Wilcoxon signed-rank tests, with a global Benjamini–Hochberg FDR correction across the analysis. No displayed comparison remained significant after correction ($p_{adj} < 0.05$).

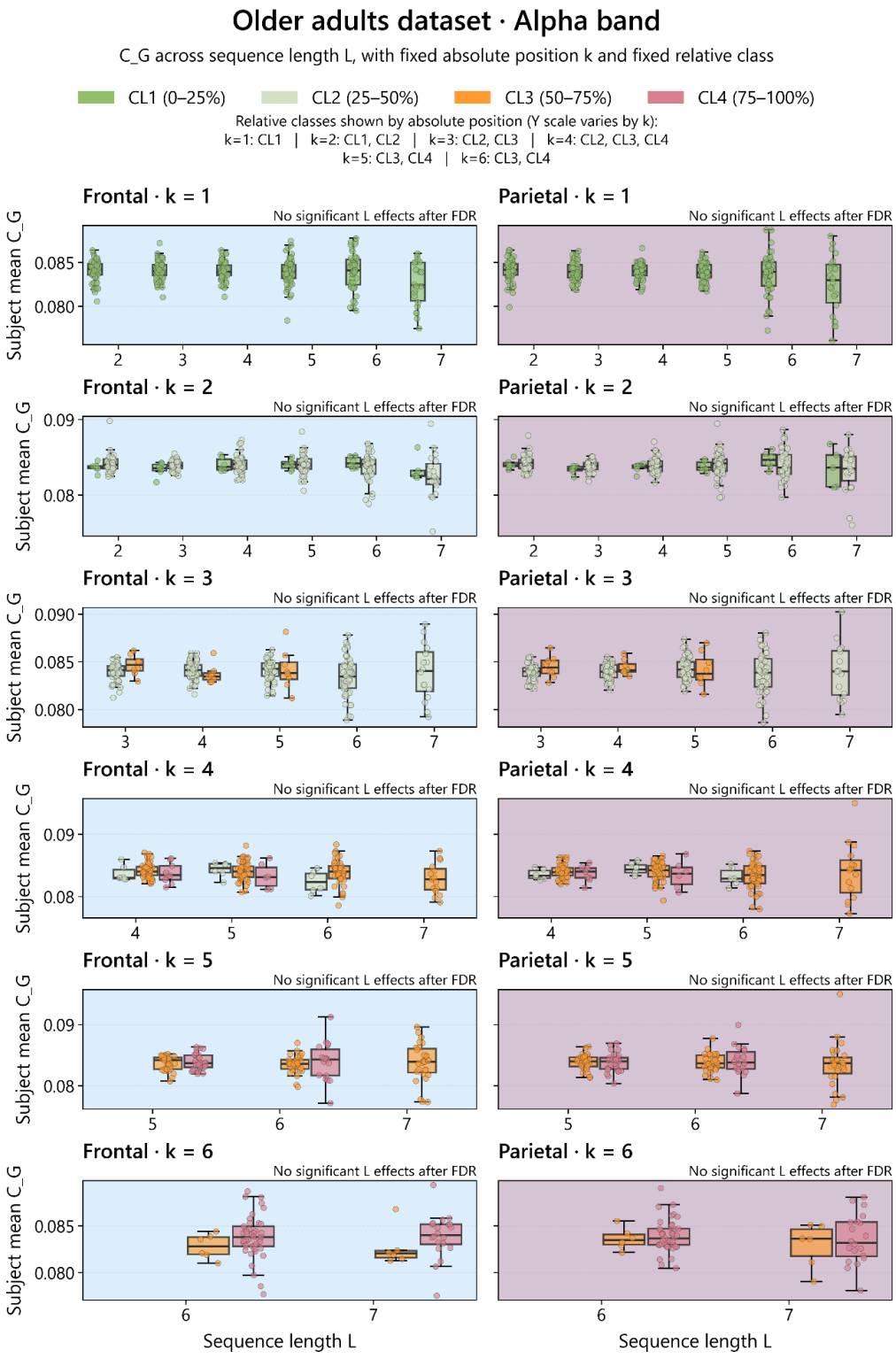


Fig. S12. Shannon graph complexity across sequence lengths in the older-adult dataset (alpha band, 8–13 Hz). Rows represent absolute serial positions (k = 1–6); columns show frontal (left) and parietal (right) results. The horizontal axis indicates sequence length (L), and colors identify relative classes CL1–CL4, defined using each participant's Corsi span. Each point represents a participant's mean C_G computed from clean epochs belonging to correctly completed trials. Boxes indicate the interquartile range and central lines the median. Vertical scales are shared within rows and vary between rows. Sequence lengths were compared within each fixed k and class using paired Wilcoxon signed-rank tests, with a global Benjamini–Hochberg FDR correction across the analysis. No displayed comparison remained significant after correction ($p_{adj} < 0.05$).

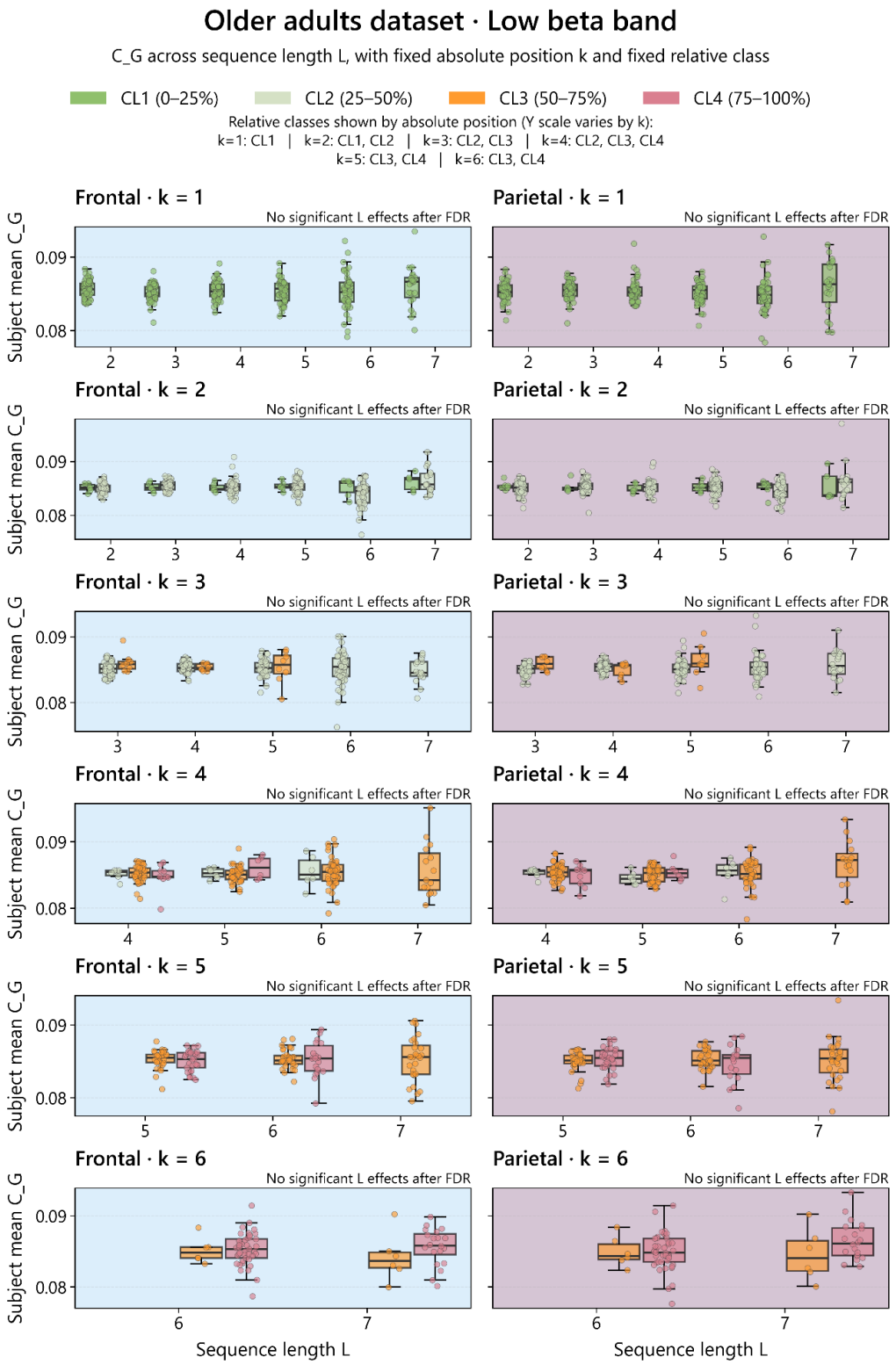


Fig. S13. Shannon graph complexity across sequence lengths in the older-adult dataset (low beta band, 13–20 Hz). Rows represent absolute serial positions (k = 1–6); columns show frontal (left) and parietal (right) results. The horizontal axis indicates sequence length (L), and colors identify relative classes CL1–CL4, defined using each participant's Corsi span. Each point represents a participant's mean C_G computed from clean epochs belonging to correctly completed trials. Boxes indicate the interquartile range and central lines the median. Vertical scales are shared within rows and vary between rows. Sequence lengths were compared within each fixed k and class using paired Wilcoxon signed-rank tests, with a global Benjamini–Hochberg FDR correction across the analysis. No displayed comparison remained significant after correction ($p_{adj} < 0.05$).

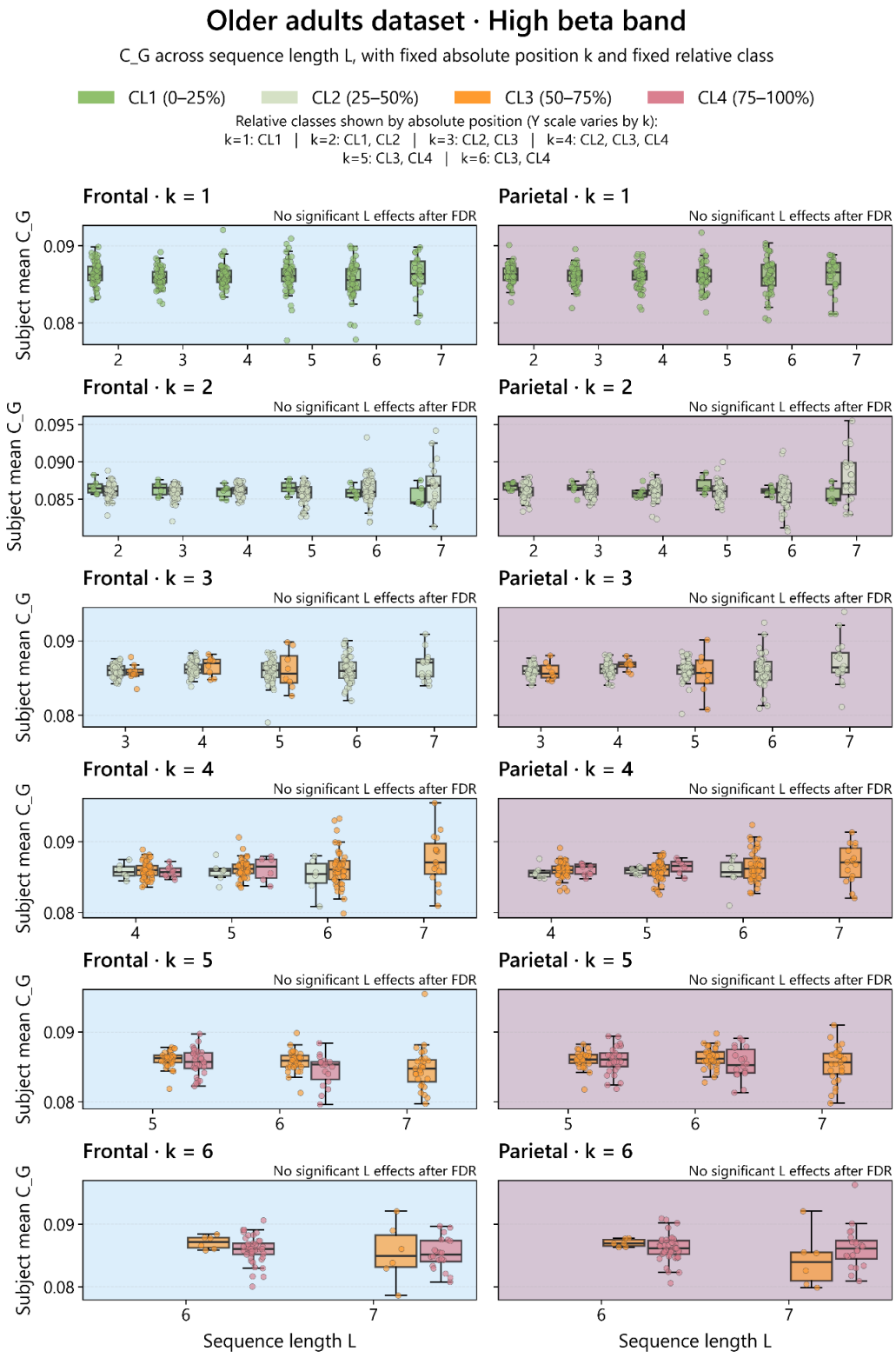


Fig. S14. Shannon graph complexity across sequence lengths in the older-adult dataset (high beta band, 20-30 Hz). Rows represent absolute serial positions (k = 1–6); columns show frontal (left) and parietal (right) results. The horizontal axis indicates sequence length (L), and colors identify relative classes CL1–CL4, defined using each participant's Corsi span. Each point represents a participant's mean C_G computed from clean epochs belonging to correctly completed trials. Boxes indicate the interquartile range and central lines the median. Vertical scales are shared within rows and vary between rows. Sequence lengths were compared within each fixed k and class using paired Wilcoxon signed-rank tests, with a global Benjamini–Hochberg FDR correction across the analysis. No displayed comparison remained significant after correction ($p_{adj} < 0.05$).

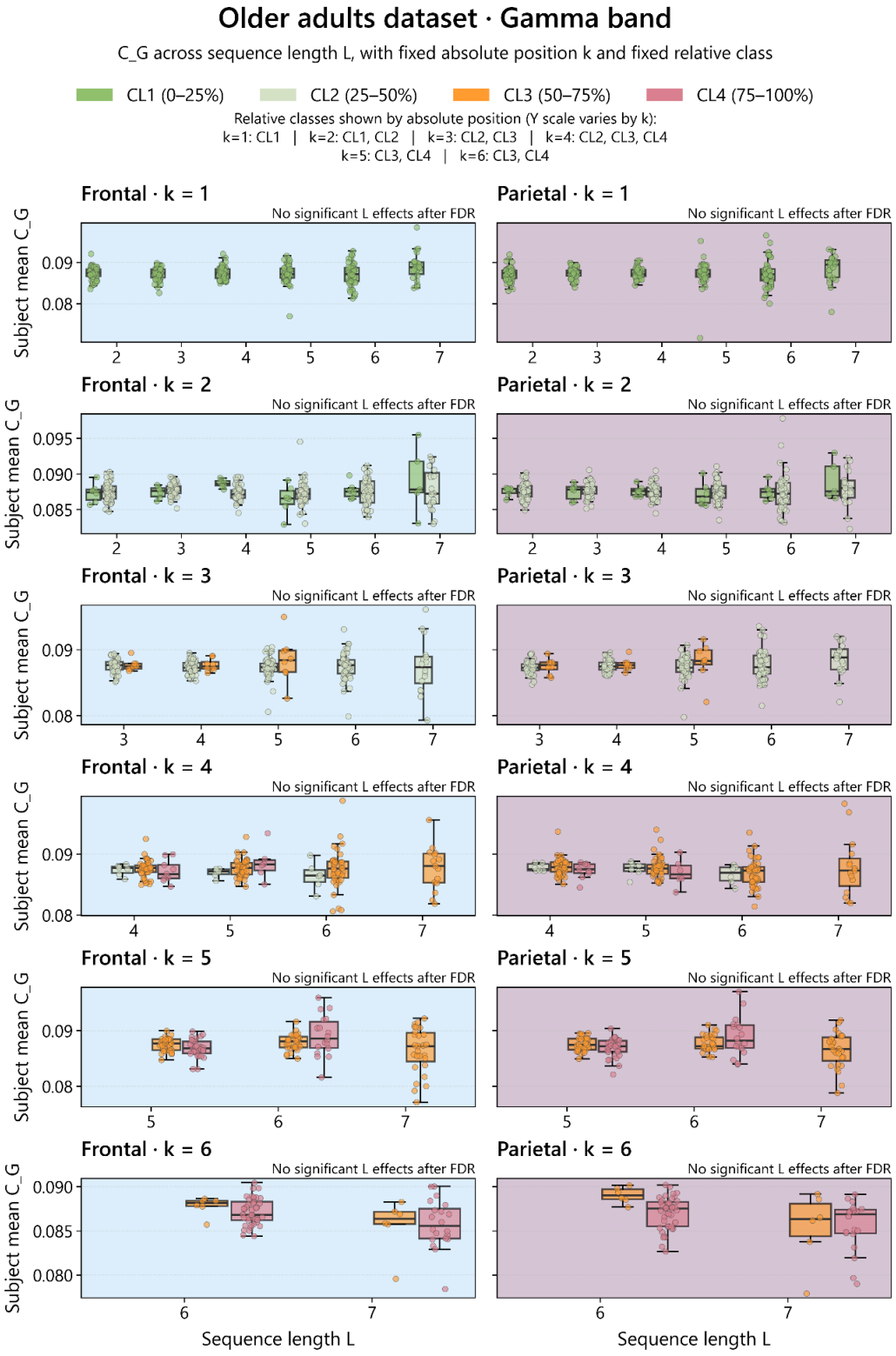


Fig. S15. Shannon graph complexity across sequence lengths in the older-adult dataset (gamma band, >30 Hz). Rows represent absolute serial positions (k = 1–6); columns show frontal (left) and parietal (right) results. The horizontal axis indicates sequence length (L), and colors identify relative classes CL1–CL4, defined using each participant's Corsi span. Each point represents a participant's mean C_G computed from clean epochs belonging to correctly completed trials. Boxes indicate the interquartile range and central lines the median. Vertical scales are shared within rows and vary between rows. Sequence lengths were compared within each fixed k and class using paired Wilcoxon signed-rank tests, with a global Benjamini–Hochberg FDR correction across the analysis. No displayed comparison remained significant after correction ($p_{adj} < 0.05$).